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Report No: 145392-MV

INTERNATIONAL DEVELOPMENT ASSOCIATION

SUPPLEMENT TO THE
PROJECT APPRAISAL DOCUMENT
ON A RESTRUCTURING AND A
PROPOSED GUARANTEE
IN THE AMOUNT OF UP TO US\$ 5 MILLION
TO
THE REPUBLIC OF MALDIVES
FOR THE
ACCELERATING SUSTAINABLE PRIVATE INVESTMENTS IN RENEWABLE ENERGY (ASPIRE) PROJECT

January 23, 2020

Energy & Extractives Global Practice
South Asia Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective Dec 31, 2019)

Currency Unit = Maldivian Rufiyaa (MVR)

MVR 15.46 = US\$1

FISCAL YEAR

January 1 - December 31

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ABBREVIATIONS AND ACRONYMS

ARISE	Accelerating Renewable Energy Integration and Sustainable Energy
ASPIRE	Accelerating Sustainable Private Investments in Renewable Energy
CPF	Country Partnership Framework
EPA	Environmental Protection Agency
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESRS	Environmental and Social Review Summary
GDP	Gross Domestic Production
GoM	Government of Maldives
GRS	Grievance Redress Service
IDA	International Development Association
IPP	Independent Power Producer
IUFR	Interim Unaudited Financial Report
ME	Ministry of Environment
MFD	Maximizing Finance for Development
MWSC	Male' Water and Sewerage Company
NDC	Nationally Determined Contribution
OP	World Bank's Operational Policy
ORAF	Operation Risk Assessment Framework
PAD	Project Appraisal Document
PDO	Project Development Objective(s)
PMU	Project Management Unit
PPA	Power Purchase Agreement
PQ	Pre-qualification
PS	Performance Standards
SREP	Scaling-up Renewable Energy Program
STELCO	State Electric Company Limited
STO	State Trading Organization
WB	World Bank



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DATASHEET

BASIC INFORMATION

Country(ies)	Project Name	
Maldives	Accelerating Sustainable Private Investments in Renewable Energy	
Project ID	Financing Instrument	Environmental Assessment Category
P145482	Investment Project Financing	B-Partial Assessment

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☐ Contingent Emergency Response Component (CERC)
☒ Series of Projects (SOP)	☐ Fragile State(s)
☐ Disbursement-linked Indicators (DLIs)	☒ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☒ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternate Procurement Arrangements (APA)	

Expected Approval Date	Expected Closing Date
14-Feb-2020	30-Sep-2021

Bank/IFC Collaboration

No

Proposed Development Objective(s)

The project development objective of ASPIRE Project is to increase PV generation in Maldives through private sector investment.

Components

Component Name	Cost (US\$, millions)
Technical Assistance (TA) to Government of Maldives	1.75



Structuring and Delivery of Tariff Buy Down for Currently Planned and Subsequent Projects	6.034
Security Package	19.9

Organizations

Borrower:	Republic of Maldives
Implementing Agency:	Ministry of Environment (ME)

PROJECT FINANCING DATA (US\$, Millions)

SUMMARY

Total Project Cost	27.684
Total Financing	27.684
of which IBRD/IDA	16.00
Financing Gap	0.00

DETAILS

World Bank Group Financing

International Development Association (IDA)	16.00
Guarantee	16.00

INSTITUTIONAL DATA

Practice Area (Lead)

Energy & Extractives

Contributing Practice Areas

Infrastructure, PPP's & Guarantees

SYSTEMATIC OPERATIONS RISK-RATING TOOL (SORT)

Risk Category	Rating
1. Political and Governance	● Moderate
2. Macroeconomic	● Substantial



3. Sector Strategies and Policies	● Moderate
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Moderate
6. Fiduciary	● Moderate
7. Environment and Social	● Moderate
8. Stakeholders	● Moderate
9. Other	
10. Overall	● Moderate

COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

Safeguard Policies Triggered by the Project	Yes	No
Environmental Assessment OP/BP 4.01	✓	
Performance Standards for Private Sector Activities OP/BP 4.03	✓	
Natural Habitats OP/BP 4.04	✓	
Forests OP/BP 4.36		✓
Pest Management OP 4.09		✓
Physical Cultural Resources OP/BP 4.11	✓	
Indigenous Peoples OP/BP 4.10		✓
Involuntary Resettlement OP/BP 4.12		✓
Safety of Dams OP/BP 4.37		✓
Projects on International Waterways OP/BP 7.50		✓
Projects in Disputed Areas OP/BP 7.60		✓



Legal Covenants

Sections and Description

The Government of Maldives shall maintain, throughout the implementation of the Project, a Project Management Unit within MEE, responsible for implementation of the Project.

Sections and Description

The Recipient shall: (a) notify the World Bank of any downgrade in the credit ratings of the escrow bank; (b) ensure that the escrow account(s) shall at all times contain funds at least equal to the total amount of Securities issued; (c) ensure that the escrow account(s) are replenished in the event of a draw down; (d) notify the World Bank in the event of a draw down on an escrow account; and (e) ensure that proceeds from the grant transferred to the escrow account are protected against set-off, seizure, or attachment.

Sections and Description

In the event that any of the proceeds of the grant allocated to Securities will not have been committed to a security by the Closing Date, then Recipient and the World Bank shall consult and agree on the use of such proceeds by the Closing Date, failing which the Recipient shall refund all such uncommitted proceeds, and the World Bank shall cancel all of such proceeds.

Sections and Description

In the event that any proceeds of the grant have been committed to Securities and will remain so committed as of the Closing Date, then the proceeds shall be retained in the escrow account(s) established for such securities until such securities have expired/terminated subject to oversight arrangements consistent with those required by the Project Operations Manual, and thereafter, the proceeds may be transferred from the escrow account(s) to the Recipient to be used for productive purposes.

Sections and Description

The Recipient shall carry out the Project in accordance with the Project Operations Manual, including the bidding and selection of subprojects, and the provision of Tariff Buy Downs and/or Securities.

Sections and Description

Prior to making any Securities available to investors, the Recipient shall enter into one or more Escrow Agreements with an escrow bank acceptable to the World Bank, for the establishment and maintenance of one or more escrow accounts from which claims by investors under Securities may be paid; such Escrow Agreement(s) to be in form and substance satisfactory to the World Bank, and meeting the requirements set forth in the Grant Agreement.

Sections and Description

For each subproject to be supported through a Tariff Buy Down or a Security, the Recipient shall enter into an Implementation Agreement with the investor, meeting the requirements of the Grant Agreement, in form and substance satisfactory to the World Bank, and containing the terms and conditions set forth in the Grant



Agreement with respect to the provision of Tariff Buy Downs or Securities.

Sections and Description

The Recipient shall ensure that, by no later than June 30, 2015, at least one subproject has been identified and the relevant Implementation Agreement and PPA have been entered into with the respective investor, all in form and substance satisfactory to the World Bank.

Sections and Description

The Recipient shall ensure that the Project and subprojects are carried out in accordance with the provisions of the Safeguards Instruments (ESMF, EIAs, and EMPs), and the World Bank Performance Standards.

Sections and Description

Covenants and Conditions for the World Bank Guarantees would be agreed when the guarantee related agreements are finalized with the Government of Maldives, STELCO/FENAKA and subproject investors.

Conditions

Type Disbursement	Description No withdrawal of Grant proceeds can be made prior to the date of execution of the Grant Agreement, and until the Recipient has finalized and adopted the Project Operations Manual (POM) in form and substance satisfactory to the World Bank.
Type Disbursement	Description The first disbursement with respect to a Tariff Buy Down will be conditioned upon the following: (i) the investor's entry into a Power Purchase Agreement (PPA) in form and substance satisfactory to the World Bank, (ii) signing of an Implementation Agreement on the subproject, in form and substance satisfactory to the World Bank, and (iii) fulfillment of other disbursement conditions specified in the POM for release of Tariff Buy Down (Component 2) payments. Subsequent claims against Component 2 will be processed by the Bank upon submission of a withdrawal application which shall include the Bank's no-objection letter for subprojects as an attachment, as stipulated in the disbursement letter.
Type Disbursement	Description The full amount allocated to Component 3 will be disbursed as advance once the following conditions are met: a. At least one Implementation Agreement in form and substance satisfactory to the Bank has been entered between the GoM and an investor for purposes of a security. b. At least one PPA in form and substance satisfactory to the Bank has been entered into between the STELCO/FENAKA and an investor in a subproject. c. GoM and the Bank have reached agreement on all the terms of the escrow mechanism,



- including the determination of the risks to be covered by GoM under such mechanism.
- d. An Escrow Agreement between the GoM and a commercial bank has been entered with respect to a related Escrow Account, in form and substance satisfactory to the World Bank.
 - e. All legal opinions or certificates required by the World Bank attesting that the Escrow Agreement has been duly executed, delivered and is legally binding on the parties thereto have been provided.
-



Supplement to Project Appraisal Document¹
Maldives Accelerating Sustainable Private Investments in Renewable Energy (ASPIRE) Project

I. INTRODUCTION

1. This Supplement to Project Appraisal Document (PAD) seeks approval from the Executive Directors to issue a guarantee in an amount of up to US\$5 million² to provide partial termination coverage for a Power Purchase Agreement (PPA) to be entered into between State Electric Company Limited (STELCO) and Ensys Company Limited.

2. The Maldives Accelerating Sustainable Private Investments in Renewable Energy (ASPIRE) Project (the “Project”), which is supported by SDR 10.325 million (US\$16 million equivalent) in guarantees in aggregate from the International Development Association (IDA) and a US\$11.684 million grant from the Scaling-up Renewable Energy Program (SREP) of the Climate Investment Funds, was approved by the World Bank Board of Executive Directors on June 26, 2014. The guarantee of up to US\$5 million requested in this Supplement to PAD is part of the series of guarantees in an amount of US\$16 million under the ASPIRE Project.

3. At the time of appraisal, a proposed series of IDA guarantees (aggregate US\$16 million) was presented to the Board, and it was agreed that any subsequent transaction in the series would be presented to the Board on absence-of-objection basis once the transactions were ready and the due diligence and safeguard requirements were completed. The issuance of a first IDA guarantee was appraised as part of the Board-approved package in 2014 in the amount of US\$2.5 million and is now effective as part of the ASPIRE guarantee series.

4. In connection with the foregoing, this Supplement to PAD has been prepared to support the installation of a 5 MW Grid-tied Solar Photovoltaic System in the Greater Malé Region on a design, build, finance, own, operate, and transfer basis. IDA intends to support this 5 MW Grid-tied Solar Photovoltaic System by issuing a second guarantee in the amount of up to US\$5 million to provide partial termination coverage for a PPA to be entered into between STELCO, the state-owned offtaker and Ensys Company Limited, the private independent power producer.

II. STRATEGIC CONTEXT

A. Country Context

5. Maldives is an island state comprising nearly 1,200 coral islands grouped into 26 atolls, spread across roughly 90,000 square kilometers of Indian Ocean. The Maldivian population, about 515,696 as of 2018, is widely dispersed across the islands, many of them remote, and physically vulnerable to rising sea levels. Eighty percent of the total land area of the country, which is less than 300 square kilometers, is lower than 1 meter above mean sea level. The country’s exposure to natural hazards and climate variability poses a threat to lives and the economy. More than 30 percent of the population live in the capital city Malé, while the rest are distributed among just under 200 other inhabited islands. Basic human development indicators are high. Maldives ranks 101 out of 189 countries in the Human Development Index (HDI) for 2017, the second-highest HDI rank in South Asia after Sri Lanka.

¹ Project Appraisal Document No: 84835-MV dated May 23, 2014

² The exact amount to be confirmed upon negotiations with the guarantee beneficiary



6. Real GDP grew by 6.7 percent in 2018 and is a middle-income country with gross domestic production (GDP) per capita of US\$10,224 (2018). The sustained growth performance has translated into significant gains in poverty reduction and Maldives performs well on poverty outcomes compared to its regional, income, and small island peers. The economy is dependent on a small number of sectors, with the following having the largest share to GDP in 2018: tourism (20.2 percent), transport and communication (11.1 percent), construction (9.3 percent), and trade (9.0 percent). The high share of tourism in the economy is both a strength and a limitation. The rapid rise in economic standards and living conditions in Maldives over the last three decades has been driven by fast growth in tourism. However, it also makes the Maldivian economy highly vulnerable to fluctuations in global economic through their effects on tourism and the direct and indirect transmission of these effects to other sectors. Periods of global recession, when tourist arrivals have fallen, have been particularly difficult for Maldives. The high dependence on tourism, and its economic benefits and risks, are typical of small island economies.

7. The extremely dispersed population has led to a high cost of service delivery. The government is emphasizing decentralization and the development of the atolls to facilitate improved service delivery for citizens. The government is seeking to improve the efficiency and effectiveness of public services, promoting the development of regional hubs where necessary, while also exploiting economies of scale and scope in larger urban areas to enable agglomeration benefits. Under this strategy, the government is developing infrastructure, housing and public services, and promoting economic development on the larger islands.

8. In recent years, the government has embarked in an ambitious infrastructure program. These upfront investments – to address bottlenecks in the tourism sector, increase resilience to climate change, ease constraints in service delivery and support housing projects— have led to a rapid increase in public and publicly guaranteed external debt. Maldives' total debt is estimated at 75.3 percent in June 2019. Debt-to-GDP grew by 10.9 percentage points between end-2017 and June-2019, mainly on account of a large leap in guaranteed debt – which rose from 4.3 percent of GDP to 16.3 percent. Maldives' risk of external debt distress remains high (IMF, 2019).

B. Sectoral and Institutional Context

9. **While the Maldives achieved universal electrification in 2008, at 14-50 US cents/kWh³ it has one of the highest residential end-user tariffs in South and South-East Asia.** The total installed generation capacity of the inhabited islands is about 247 MW, with the resort islands having an additional 144 MW (which are managed independently of the government) and 20 MW on industrial islands. Total annual electricity consumption was 704 GWh in 2017, about 56 percent of which is accounted for by the Greater Malé region. Electricity consumption per capita was 1840 kWh/year in 2016 and electricity demand is projected to grow 7 percent per annum.

10. **The Maldives comprises a number of small, isolated island-based grid systems that pose challenges for security of supply and high costs of electricity service.** Three vertically integrated state-owned utilities STELCO, FENAKA, and MWSC⁴ operate a total of 186 powerhouses on inhabited and industrial islands. STELCO is the utility serving Male, while FENAKA was formed in 2012 as a merger of 6 regional utilities to serve the outer islands, except resort islands.

11. **The sector is almost 95 percent reliant on diesel fuel for power generation and diesel imports range from US\$240-400 million annually, resulting in budget uncertainty and fiscal burden for the Government.** End-user

³ ME, "Ministry of Environment, Maldives - Island Electricity Data Book," Male, 2018

⁴ MWSC operates 1 power house on V.Rakeedhoo.



tariffs are insufficient to cover costs and, despite significant budgetary support being provided to utilities by the GoM, the electricity sector is not financially sustainable. Costs for the most efficient diesel generator in the Maldives are estimated to range between 23-33 US cents/kWh, while for many of the outer islands costs can be as high as 70 US cents/kWh. Energy demands have grown exponentially over the past three to four decades as a result of unprecedented economic growth in key sectors and increased demand for electricity generation. The demand for energy generation is still expected to grow fast, increasing pressure on the fiscal budget and balance of payments.

12. **Subsidies paid by the Government of Maldives (GoM) to the electricity sector reached about US\$60 million in 2018, about 1 percent of GDP.** Both STELCO and FENAKA rely heavily on GoM subsidies. Fuel subsidy accounts for the volatility in fuel prices and makes up about 65 percent of the direct subsidy budget in 2018. It is applied if the fuel cost per liter is above a threshold baseline rate. The difference between the market price and baseline is then paid off directly to State Trading Organization (STO) as a subsidy to allow the utilities to purchase fuel from STO at a lower price. Usage subsidy accounts for about 35% of latest direct subsidy budget, to be paid to the utilities to compensate an adjustment for harmonizing domestic and business category tariffs across all islands.

13. **Increasing renewable energy penetration in the electricity sector is a solution to address the current challenges.** The cost of generation is lower from renewable energy than from diesel. It would not only improve financial performance of the public utilities but also reduce the need for budget support for the electricity sector, thus improving fiscal sustainability. Furthermore, renewable energy would positively impact the cost of electricity service as logistical difficulty of shipping and storing fuel across a large number of islands would decrease. Therefore, it would enhance electricity service reliability and security as well as resilience against natural disasters.

14. **The Government of Maldives is committed to harnessing renewable energy resource and increasing its use.** Reducing dependence on imported fuel and investing in renewable energy is a key priority of the Government. Peaceful transition of power after the September 2018 elections has ushered in a new Government that is prioritizing resilience and sustainability of the country. The Maldives Energy Policy and Strategy 2016 seeks to promote renewable energy in the country and to encourage private sector renewable energy development as one of its nine key policies. Moreover, the national Strategic Action Plan for the Maldives (2019-2023) includes a specific pillar for Clean Energy with clear renewable energy targets to increase the share of renewable energy by 20% compared to 2018 levels. Reinforcing the commitments made under the SAP, the Minister of Environment announced a target to increase the share of renewable energy to 70 percent by 2030 at the UN Climate Action Summit in September 2019.

15. **To support renewable energy development, the World Bank has been engaging with the Government through the ASPIRE project.** The Bank-financed ASPIRE project has been under implementation since its approval in June 2014. Implementation has been slow, as the project for the first time in the Maldives attempted to bring in private sector investment in rooftop PV in the service areas of public utilities using IDA guarantee and there were issues of quality of rooftops available to mount the panels. The first sub-project of 1.5 MW of rooftop solar PV in Hulhumale' has been operational since March 2018. A private sector developer was awarded through an international competitive tendering to be an independent power producer (IPP). It sells electricity generated from the rooftop solar PV to STELCO under a 20-year power purchase agreement (PPA) at a fixed price of 21 US cents/kWh, which is lower than the all-in cost of diesel-based power generation in the Maldives. US\$3.3 million of private capital has been mobilized for the first sub-project.

C. Relevance to Higher Level Objectives



16. **The ASPIRE Project supports the following two broad engagement areas of the Maldives Revised Country Partnership Framework (CPF):** (i) Pillar 2 on Strengthening Natural Resource Management and Climate Resilience – The Project aims to support electricity generation from RE and improve climate resilience by diversifying energy mix; and (ii) Pillar 3 on Strengthening Fiscal Sustainability – This Project aims to contribute to strengthening fiscal sustainability by reducing fossil fuel import and thus government subsidy to fossil fuel and electricity.
17. **The Project is closely aligned with Government’s policies and national plans** including the Nationally Determined Contribution (NDCs) targets, the national Strategic Action Plan (2019-2023) and the announcement by the Minister of Environment to target to increase the share of renewable energy to 70 percent of the generation mix in the country by 2030.
18. **Maximizing Finance for Development (MFD), the Project was able to successfully bid out the first sub-project.** The first sub-project attracted interests from the private sector and mobilized private investment of US\$3.3 million. Investment climate for renewable energy in the Maldives has been improved through developing a full package of financeable contracts and building local capacity in implementing such a sub-project. Supporting subsequent sub-projects is expected to further increase private capital mobilized for renewable energy.

III. PROJECT DESCRIPTION

19. **Project Development Objective(s) (PDO):** The PDO of ASPIRE⁵ to increase PV generation in Maldives through private sector investment remains unchanged.
20. **Components.** There is no proposed change in the Project components. The Project therefore continues to implement three components: (1) technical assistance support to GoM, (2) structuring and delivery of tariff buy downs for currently planned and subsequent subprojects, and (3) a security package consisting of: (i) escrow accounts to cover payment defaults by the offtaker under the subproject power purchase agreements (PPAs), and (ii) IDA guarantees to partially backstop the payment by the Government of subproject termination payments. IDA funding is used for this guarantee support only, covering partial termination payments as part of the security package under Component 3. Grant from the Scaling-up Renewable Energy Program (SREP) is used for all other components, including to fund escrow accounts as part of the security package under Component 3 to be used for short-term payment shortfalls. US\$16 million of IDA guarantee and US\$11.684 million of SREP grant was approved for the three components on June 26, 2014 by the Board of Executive Directors.
21. **The first sub-project has proven the private-sector oriented approach to developing renewable energy in the Maldives.** The first sub-project is fully compliant with all World Bank requirements, and even though it was implemented on the roofs of already occupied social housing units, the Grievance Redress Mechanism has not received a single complaint against the Project during construction, or since. It is also pertinent to note that while there were nearly two years of delays (from contract award to commencement of construction) for various reasons, the actual construction of the facility was very smooth and took less than six months. The first sub-project successfully navigated these issues and has been selling sustainable solar PV generated electricity to STELCO under a 20-year PPA and backed by a security package that includes an IDA partial risk guarantee, since March 2018.

⁵ Page 7, para 22 of the Original PAD.



22. **Building upon the contracts developed and lessons learned from the first sub-project, the second sub-project was developed.** Multiple sites for a total of 5 MW of potential installed capacity were identified in Greater Male'. The sub-project is comprised of rooftops of Government-owned buildings and spaces on the side of the road connecting Hulhumale' and the international airport in the North Male' Atoll. Canopies are proposed to be installed on the road sides to install solar PV and allow land sites to be available for other uses. Hence no land acquisition is required. The bid package includes a security package of SREP grant for the escrow mechanism and IDA guarantee from Component 3 as well as US\$2 million of Tariff Buydown Grant from Component 2 to lower tariff bids while keeping the level of risk mitigation same with the first sub-project.

23. **The second sub-project was competitively tendered in 2019, and its results indicate that the investment climate in solar PV by the private sector has been significantly improved.** Pre-qualification (PQ) tender received 34 registrations from Maldivian, regional and global companies; 21 of these filed completed applications, which was a significant improvement over the first sub-project. Of these, 16 were deemed qualified by GoM, thus laying a firm foundation for renewable energy investments in the Maldives and opening a new field for foreign investments. On September 12, 2019, bids were opened, and six companies submitted bids. The lowest bid was 10.9 US cents per kWh for a 15-year PPA, significantly lower than 21 cents per kWh for the first sub-project thanks to lower solar costs, the security package (IDA guarantee covering partial termination + SREP grant funding the escrow account) proposed in accordance with the Project design as well as the use of the Tariff Buydown (not used in Phase I).

24. The PPA will be entered into between STELCO and Ensys Company Limited, the private independent power producer. Established in 1996, Ensys Company Limited is a company incorporated and existing under the laws of Thailand with ample experience in grid-connected solar PV ranging up to 20 MW scale in Thailand⁶. The solar PV facility of the second sub-project is expected to generate 7,870 MWh in the first year of operation with 16 percent of capacity utilization factor. Timely implementation of the second sub-project will be key in accelerating RE investments in the Maldives and opening a new field for investments and employment. GoM is expected to sign the PPA with the selected bidder in February 2020.

25. The updated disbursements for the components are shown in Table 1 below. The original allocated amounts are the numbers without brackets. The numbers within brackets represent the amounts disbursed or committed (in the case of the IDA guarantee) within that component, per the Project's Interim Unaudited Financial Report (IUFR) for the third quarter of 2019 (July - September 2019).

Table 1: Updated Project Disbursements as of December 2019 (in US\$ million)

<i>ASPIRE Component</i>	<i>SREP</i>	<i>IDA Guarantee</i>
Component 1: Technical Assistance Support	1.75 (1.07)	0 (0)
Component 2: Structuring Tariff Buydown	6.03 (0)	0 (0)
Component 3: Security Package	3.9 (0.24) *	16.0 (2.50) #
Total	11.68 (1.31)	16.0 (2.50)

* This is earmarked for the escrow security mechanism

This is earmarked for partial risk guarantees to be issued for sub-projects

26. The second sub-project is projected to utilize US\$2 million of Tariff Buydown grant under Component 2, about US\$0.5 million of SREP grant for the escrow mechanism and up to US\$5 million of IDA guarantee under Component 3, subject to finalization of negotiations with the winning bidder. Supporting the second sub-project

⁶ Note that certain integrity checks have been completed (Web search; INT Company Risk Profile Database; list of debarred firms) and will be supplemented with integrity due diligence from IFC and/or MIGA to the extent involved.



would expedite the disbursement of SREP grant and commitment of the Bank guarantee under the Project and enable the Borrower to enhance the development effectiveness of the operation for the Project.

27. In November 2019, the closing date of the Project has been extended by 21 months from December 31, 2019 to September 30, 2021. The extension was granted to support the second sub-project in accordance with the bidding package that indicates ASPIRE's financial support. It has allowed the Borrower to provide positive signals to the market regarding the credibility of the GoM to run a competitive tendering process to international standards and result in an improved enabling environment for future private sector investment in renewable energy.

IV. IMPLEMENTATION

A. Institutional and Implementation Arrangements

28. **Implementing Agencies.** ASPIRE has been managed through a Project Management Unit (PMU) at GoM's Ministry of Environment (ME). ME has been collaborating in active World Bank projects such as the Maldives Clean Energy for Climate Mitigation Project (P128268), Maldives Environmental Management Project (P108078), and Maldives Ari Atoll Solid Waste Management Project (P130163). Additionally, ME has successfully implemented the first ASPIRE subproject (1.5 MW Hulhumale'). Having learnt valuable lessons from this implementation, ASPIRE's PMU has added a lawyer, environmental and social safeguards specialist, and a project engineer to its ranks. This allows ME to handle much of the sub-project preparation work in-house – something that will become increasingly important with the development of renewables in the Maldives.

B. Results Monitoring and Evaluation Arrangements

29. Overall monitoring and evaluation of the Project will remain the responsibility of ME with no change. ME will continue to carry out monitoring and evaluation of the Project components and sub-projects supported under the Project in accordance with the indicators included in the original results framework.

V. APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis

30. Incorporating lessons from the first sub-project, the Bank conducted a due diligence of the second sub-project. The bidding package for the second sub-project, including all project documents such as the PPA, Implementation Agreement, Roof Lease Agreement, Escrow Agreement, and Licensing Agreement, has been revised for the second sub-project with lessons incorporated from the first sub-project. Contracts for the second sub-project also include appropriate provisions for a tariff buy-down, to keep bid prices in an acceptable range. The lowest bid of 10.9 US cents per kWh is below the utility's *subsidized* variable cost of diesel-generation, resulting in savings at the utility and national levels.

31. STELCO, the utility covering the Greater Male region is the offtaker for the second sub-project. The utility, which benefits from a fuel subsidy which caps STELCO's fuel costs at a certain level independently from the market price, has been profitable over the last two years in 2016 and 2017. Additionally, STELCO has been setting a positive



track record for the second sub-project. Nevertheless, STELCO is faced with upcoming investment needs for new generation capacity as well as sizeable amount of trade payables on its balance sheet which will need to be monitored.

B. Fiduciary

32. **Financial Management.** Adhering to the structure outlined in the Original PAD, the Project has complied with all the Financial Management requirements. The Project was rated Satisfactory in the latest implementation status report and as outlined in the Original PAD all the Bank projects implemented by ME have been rated as Satisfactory on the Financial Performance.

33. **Procurement.** The Project is being carried out in accordance with: World Bank "Guidelines: Procurement of Goods, Works and Non-Consulting Services under IBRD Loans and IDA Credits & Grants by World Bank Borrowers" dated January 2011 (Procurement Guidelines); "Guidelines: Selection and Employment of Consultants under IBRD Loans and IDA Credits & Grants by World Bank Borrowers" dated January 2011 (Consultant Guidelines); and the provisions stipulated in the SREP Grant Agreement. The procurement performance of the Project is rated as "satisfactory". All the bidding documents, including invitation for pre-qualification, request for bids and bid evaluation reports, have been reviewed and provided no-objection by the Bank to ensure compliance with the relevant Bank policies.

C. Safeguards

34. **Environmental Safeguards.** Since ASPIRE follows a private sector approach as per the original PAD (see paragraph 100), Performance Standards are followed for each sub-project. The first sub-project is compliant with the World Bank's Operational Policy (OP) 4.01 – Environmental Assessment and OP 4.03 (including Performance Standards (PS) 1: Assessment and management of environmental and social risks and impacts, 2: Labor and working conditions; 3: Resource efficiency and pollution prevention and 4: Community health, safety and security). Therefore, the environmental safeguards performance is rated as 'Satisfactory'. The overall safeguard risk continues to be "Moderate."

35. **Social Safeguards.** Activities under the first sub-project have not led to any adverse social impact on the community in terms of land acquisition and resettlement. Awareness-raising activities among relevant community members, and consultations with the stakeholders prior to the installation of solar PVs, have been carried out regularly as per the Environmental and Social Management Framework (ESMF). Social safeguards performance is thus rated as 'satisfactory.'

36. The second sub-project has been also following the relevant Bank safeguards policies. No revision has been made in the ESMF, as there is no additional risk anticipated for the second sub-project or the need to trigger additional PSs. As per the request of the Environmental Protection Agency (EPA) and Civil Aviation a specific Glare Assessment has also been conducted and cleared and has confirmed that the impacts to flight operations will be almost negligible. Although the sub-project utilizes spaces on the side of the road, canopies will be built for installing solar PV therefore no land acquisition is required.

37. In accordance with the ESMF, ME submitted during the bid evaluation period the Environmental and Social Management System and the Independent Environmental Certification of Ensys Company Limited, the winning bidder. The Bank has reviewed them to assess the environmental and social performance and compliance standards



of the winning bidder and provided clearance. The requisite clearances (e.g., civil aviation, HDC, Ministry of Defense) have been obtained for the installation of solar panels on the side of the link road under the second sub-project. Following procedures set out in the ESMF, once the contract is awarded, expected in February 2020, Environmental and Social Management Plans (ESMPs) will be prepared by Ensys Company Limited, review by the Bank and EPA, and disclosed to public. The Bank will prepare the Environmental and Social Review Summary (ESRS) in accordance with OP 4.03 upon review of ESMPs.

38. **Grievance Redress Services.** Communities and individuals who believe that they are adversely affected by a World Bank (WB) supported project may submit complaints to existing project-level grievance redress mechanisms or the WB's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the WB's independent Inspection Panel which determines whether harm occurred, or could occur, as a result of WB non-compliance with its policies and procedures. Complaints may be submitted at any time after concerns have been brought directly to the World Bank's attention, and Bank Management has been given an opportunity to respond. For information on how to submit complaints to the World Bank's corporate Grievance Redress Service (GRS), please visit <http://www.worldbank.org/en/projects-operations/products-and-services/grievance-redress-service>. For information on how to submit complaints to the World Bank Inspection Panel, please visit www.inspectionpanel.org.

39. The first sub-project is compliant with World Bank procedures for the establishment of a Grievance Redress Mechanism. As noted in paragraph 7, for the first sub-project, no complaint was received during construction or since. ME has sufficient capacity to replicate this approach for the second sub-project.

VI. KEY RISKS

40. With the benefit of 5 years of implementation, some key take-aways are as follows. Inexperience with PV technology and a volatile business and political environment in the Maldives were challenging for the first sub-project. Resistance from various sources including the off-taker resulted in significant delays. Additionally, the Government's initial reluctance to provide a foreign exchange convertibility security to the private sector created challenges. Despite this, private investment has come in due to the persistence of the Bank and ME. The successful operation of the first sub-project has built support for the PPA model and private sector investment. This will be cemented further with the 5 MW bid under the second sub-project.

VII. RESULTS FRAMEWORK

41. The Results Framework is presented below. The current status comes from the July 2019 technical mission.



Table 3: ASPIRE Project Results Matrix

<i>Indicator</i>	<i>Current Status</i>	<i>Previous Status</i>
PDO Level Results Indicators		
Cumulative Installed Generation Capacity of Solar Energy Constructed	1.5MW in operation since March 2018; 5MW Phase II Greater Male' bid to be installed through the second sub-project.	1.5MW in operation since March 2018.
Annual Electricity Output Solar Energy under the project	2,280 MWh for the first year of the first sub-project; 7,870 MWh per year to be added through the second sub-project.	2,280 MWh for the first year of the first sub-project.
Private Capital Mobilized	Approximately \$3.3 million of private capital was mobilized for the first sub-project; about \$6 million of private capital expected to be mobilized through the second sub-project.	Approximately \$3.3 million of private capital was mobilized for the first sub-project.
Annual Reduction in Bulk Power Cost in Project Areas	Based on unsubsidized cost of diesel, the expected annual saving is \$24,700. This assumes no change in capital cost of diesel generators. When the second sub-project is commissioned, the result number will increase.	Based on unsubsidized cost of diesel, the expected annual saving is \$24,700. This assumes no change in capital cost of diesel generators.
Intermediate Results Indicators		
Development of Standard Contracts and Guidelines	Completed - Contract and concepts developed in Phase I were replicated for Phase II and released	Completed and proven for 1.5 MW Phase I
Cumulative Megawatts of PV pipeline subprojects prepared	27.5 MW prepared, including Phase I, II and a new sub-project in outer islands	6.5MW prepared

42. The Project restructuring is proposed to adjust the PDO indicators and intermediate indicators as outlined in Table 4 below. These proposed changes to the results targets will reflect the development of the first and second sub-projects. Due to implementation delays at the beginning of the Project, despite the extension of the closing date granted in November 2019, the Project would only be able to support up to the second sub-project by the closing date. Subsequent sub-projects currently under development would be proposed to be supported under the new Bank operation, Accelerating Renewable Energy Integration and Sustainable Energy (ARISE) Project (P172788), which is under preparation targeting the Board approval by May 2020.



Table 4: Summary of the proposed changes in the Results Framework

Indicators	Original Target	Proposed Target	Remark
PDO Level Results Indicators			
Cumulative Installed Generation Capacity of Solar Energy Constructed (MW)	20	6.5	End target revised
Annual Electricity Output from Solar Energy under the Project (MWh/yr)	32,610.6	10,000	End target revised
Private Capital Mobilized (US\$ million)	42	9.3	End target revised
Annual Reduction in Bulk Power Cost in the Project Areas (US\$ million)	1.63	0.5	End target revised
Intermediate Results Indicators			
Development of Standard Contracts and Guidelines (Number)	7	7	No change
Cumulative Megawatts of PV pipeline subprojects prepared (MW)	35	27.5	End target revised

VIII. INDICATIVE TERMS AND CONDITIONS FOR GUARANTEE

43. A summary of the Terms and Conditions of the Agreements to be signed in the context of providing a Termination Payment IDA Guarantee, including the Indemnity Agreement to be signed with the Government of Maldives (GoM)⁷ is provided in Annex 1.

⁷ This term sheet will be applicable to each subproject.



Annex 1: ASPIRE Guarantee Term Sheet

*This term sheet contains a summary of indicative terms and conditions of proposed guarantees (each a “**Guarantee**”) by the International Development Association (“**IDA**”) for discussion purposes only and does not constitute an offer to provide a Guarantee. The provision of a Guarantee is subject, inter alia, to satisfactory appraisal by IDA of the specific sub-project (each a “**Subproject**”) proposed under the framework of the ASPIRE Project (“**Project**”), compliance with all applicable policies of the World Bank, including those related to environmental and social safeguards, review and acceptance of the ownership, management, financing structure (including in connection with shareholders, suppliers, equipment and Project design, and contracts proposed by the winning bidder), review and acceptance of project/transaction documentation by IDA, and the approval of the management and Executive Directors of IDA, as applicable, in their sole discretion. Without limiting the generality of the foregoing, IDA is highly selective with regard to the clients and beneficiaries it works with and is diligent with Know Your Customer requirements for all Project participants (equity investors, ultimate shareholders, lenders, contractors, advisors) and will undertake an appraisal of the Project and the Project Company including an assessment on these parameters.*

IDA Guarantee Agreement	
Guarantor:	International Development Association (IDA)
Guaranteed Beneficiary:	The Project Company undertaking the relevant Subproject
Guarantee Face Value:	The IDA Guarantee Face Value will not exceed US\$5 million, with the predetermined amount to be calculated based on a percentage of the maximum termination amount payable as per the provisions of the Implementation Agreement between the Project Company and the Government of Maldives (“ GoM ”) and the Power Purchase Agreement (“ PPA ”, and together with the Implementation Agreement, the “ Project Documents ”) between the Project Company and [STELCO] as buyer under the PPA (the “ Offtaker ”). ⁸
Guarantee Support:	The Guarantee would backstop the failure by the Government of Maldives (“ GoM ”) to pay the amounts due to the Beneficiary under the relevant Project Documents, following the occurrence of a Guaranteed Event (as defined below).
Guaranteed Events:	GoM’s failure to comply with its undisputed/final termination payment obligations under the Project Documents. For the avoidance of doubt, coverage under the IDA Guarantee shall not extend to any termination payment resulting from a termination of the Project Documents due to a default by the Beneficiary [or any termination event that is otherwise insurable (e.g. natural force majeure) affecting the Beneficiary].

⁸ This term sheet is based on the proposed templates of the Project Documents reviewed by IDA. The final documentation will need to be acceptable to IDA, and IDA reserves the right to modify or withdraw the proposed Guarantees, including for reasons relating to changes in the current terms of the Project Documents.



Maximum Amount:	Guaranteed	Guarantee Face Value.
Maximum Guarantee Period:		20 years from IDA Guarantee effectiveness as referred to below
Signing:		If the Guarantee-related legal agreements are not signed within [24] months following approval by the Board of Executive Directors of IDA, IDA may withdraw the offer of the Guarantee.
Exclusions, Limitation/Suspension & Termination Events:	Withholding,	Standard exclusion, withholding, limitation/suspension and termination events for transactions of this nature.
Conditions Precedent to Effectiveness of the IDA Guarantee:		Usual and customary conditions for financings of this type, including but not limited to the following: (a) Firm commitment for sufficient financing to complete the construction of the Project, including satisfactory contribution of equity; (b) Execution, delivery and effectiveness of all Project and financing agreements, in form and substance satisfactory to IDA, including the PPA, the Implementation Agreement, the Escrow Agreement for the payment security mechanism under the PPA, the applicable Site Agreement granting the Project Company the right to use the designated site(s) for purposes of its Subproject, the Indemnity Agreement and the Project Agreement; ⁹ (c) Delivery of all relevant host country environmental approvals required for the construction and operation of the Project, and compliance with all applicable World Bank requirements relating to Sanctionable Practices and environmental and social safeguards, including the World Bank Performance Standards; (d) [Effectiveness of all required insurance (to include IDA as an additional insured on third-party liability insurance);] (e) Satisfaction of all conditions precedent for first disbursement under the financing documents if applicable, save for any condition that requires the effectiveness of the Guarantee Agreement to have occurred; (f) Provision of satisfactory legal opinions; (g) Payment in full of the [Initiation Fee] and [Processing Fee], , the first installment of the [Guarantee Fee] (if applicable) [and the reimbursement of IDA's outside legal counsel expenses]; and (h) Satisfactory integrity due diligence of Project Company (and related parties) and guaranteed parties.
Subrogation:		If and to the extent IDA makes any payment under the Guarantee, IDA will be subrogated immediately to the extent of such unreimbursed payment to the Guaranteed Beneficiary's rights under the relevant PPA and under the relevant Implementation Agreement to the extent of the unpaid amount under the relevant Project Documents that gave rise to the payment under the Guarantee.

⁹ The terms of the Project Agreement may be incorporated within the Guarantee Agreement, in which case there will be no separate Project Agreement.



Claims and disputes:	If there is a dispute between the Guaranteed Beneficiary and the Offtaker or GoM as to the Offtaker's or GoM's obligation to pay or the amount of its liability, the Guarantee would be callable only in respect of amounts that the offtaker/GoM is obligated to pay, and fails to pay. For the avoidance of doubt, IDA will pay only up to the offtaker's/GOM's liability that has been determined, whether through expert determination, settlement agreement between the parties, arbitral award, or in accordance with contractual procedures acceptable to IDA, so long as such determination is final and binding (i.e., an arbitral award is not necessarily required).
Governing law:	English law or New York Law.
Indemnity Agreement	
Parties:	IDA and Maldives (the " Member Country ")
Indemnity:	The Member Country will reimburse and indemnify IDA on demand, or as IDA may otherwise direct, for all payments under the Guarantee and all losses, damages, costs, and expenses incurred by IDA relating to or arising from the Guarantee.
Covenants:	[Usual and customary covenants included in agreements between member countries and IDA.]
Remedies:	If the Member Country breaches any of its obligations under the Indemnity Agreement, IDA may suspend or cancel, in whole or in part, the rights of the Member Country to make withdrawals under any other loan, credit or grant agreement with IBRD or IDA, or any IBRD loan or IDA credit to a third party guaranteed by the Member Country, and may declare the outstanding principal and interest of any such loan or credit to be due and payable immediately. A breach by the Member Country under the Indemnity Agreement will not, however, discharge any guarantee obligations of IDA under the Guarantee.
Governing Law:	The Indemnity Agreement will follow the usual legal regime and include dispute settlement provisions customary for agreements between member countries and IDA.
Project Agreement	
Parties:	IDA and the Beneficiary
Representations and Warranties:	The Beneficiary will represent, among other standard and project-specific provisions, as of the effective date, that: (a) it is in compliance with applicable environmental laws and the applicable World Bank guidelines, environmental and social safeguard requirements, including the World Bank Performance Standards and other applicable requirements; and (b) neither it (nor its direct and indirect shareholders and any other relevant project participants, as determined by IDA), nor any of its affiliates has engaged in any Sanctionable Practices¹⁰ in connection with the Project.

¹⁰ "Sanctionable Practices" include corrupt, fraudulent, collusive, coercive, or obstructive practices, as defined in IDA's Anti-Corruption Guidelines



Covenants:	The Beneficiary will covenant, among other things, that it will: (a) comply with applicable laws, including environmental laws, and the applicable environmental and social safeguards requirements under the World Bank Performance Standards; (b) provide annual audited financial statements and other reports; (c) provide certain notices and other information to IDA; (d) provide access to the Project; (e) not engage in (or authorize or permit any affiliate or any other Person acting on its behalf to engage in) any Sanctionable Practices in connection with the Project; (f) comply with World Bank requirements relating to Sanctionable Practices regarding individuals or firms included in the World Bank Group list of firms debarred from World Bank Group-financed contracts; and (g) obtain IDA's consent prior to agreeing to any change to any transaction document which would affect the rights or obligations of IDA under the Guarantee Agreement or any other guarantee related agreement.
Payment of Fees to IDA:	Payment of fees due to IDA is the obligation of the Project Company.
Initiation Fee:	15 bps of the Guarantee Face Value (but not less than USD 100,000) ¹¹ .
Processing Fee:	50 bps of the Guarantee Face Value. ¹²
Guarantee Fee:	75 basis points per annum. The IDA Guarantee Fee is charged on that portion of the guaranteed amount that IDA has contractually committed and for which IDA has financial exposure under the guarantee (Maximum Guaranteed Amount). The Guarantee Fee must be paid [in advance semi-annually on regular payment dates]. The Guarantee will terminate in the event of nonpayment of any installment of the Guarantee Fee.
External Legal Costs:	[Reimbursement of IDA external legal counsel expenses by the Project Company.]
Cooperation Agreement	
Parties:	IDA and the Offtaker

¹¹ The Initiation Fee amount in this case will be calculated as follows: (Guarantee Face Value /US\$16 million) x US\$100,000

¹² In exceptional cases, projects can be charged over 50 bps of the guarantee amount.



Cooperation agreement:

The Offtaker will covenant, among other things, that it will:

- (a) comply with all its obligations under the transaction documents;
- (b) obtain IDA's consent prior to agreeing to any change to any transaction document which would materially affect the rights or obligations of IDA under the Guarantee Agreement or any other transaction document;
- (c) provide certain notices to IDA;
- (d) take all action necessary on its part, in accordance with and as required by the terms of the project-related agreements to which it is a party, to enable the Beneficiary to perform all of the Beneficiary's obligations under the Project Agreement, and other relevant transaction document;
- (e) cooperate with IDA and furnish to IDA all such information related to such matters as IDA shall reasonably request; and promptly inform IDA of any condition which interferes with, or threatens to interfere with, such matters; and
- (f) reimburse and indemnify IDA on demand, or as IDA may otherwise direct, for any payments under the Guarantee related to a Guaranteed Event attributable to the Offtaker.